MAJOR CYBERATTACKS ON COMPANIES

Detailed case studies | Cybersecurity and Information Technology

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Report objective: Understand real-world cyber incidents, attacker techniques, business consequences, incident response, and practical prevention methods. This report distinguishes public disclosures from estimates and does not assume that every reported intrusion had the same cause.

Executive summary

Cyberattacks against businesses range from stolen passwords and unpatched applications to ransomware and attacks through suppliers. The cases below show that the damage can include loss of customer trust, disrupted operations, legal costs, and exposure of sensitive personal information. Strong identity security, timely patching, network segmentation, logging, and tested recovery plans reduce risk but cannot guarantee prevention.

At-a-glance comparison

Company	Year	Main incident
Yahoo	2013-14	Account data theft
Equifax	2017	Unpatched web application
Marriott / Starwood	2018 disclosed	Guest records compromised
Sony Pictures	2014	Destructive intrusion and leaks
Colonial Pipeline	2021	Ransomware / fuel disruption
T-Mobile	2021	Customer data breach
MGM Resorts	2023	Social engineering / disruption
Change Healthcare	2024	Ransomware / claims disruption
Target	2013	Vendor access / payment data
Microsoft	2024 disclosed	Corporate email intrusion

1. Yahoo (2013-2014)

Attack type: Data breach / account compromise

What happened: Yahoo disclosed in 2016 that data associated with a 2013 breach affected all approximately 3 billion accounts; a separate 2014 incident affected at least 500 million accounts.

How the attack occurred: Attackers accessed account information. Yahoo reported that stolen data included names, email addresses, phone numbers, dates of birth, and hashed passwords in some cases; payment card data was not part of the affected system as described.

Impact: Enormous account exposure, password-reset actions, regulatory scrutiny, and reputational damage. The incidents also affected negotiations surrounding the sale of Yahoo's operating business.

Response and recovery: Yahoo notified users, invalidated certain security questions, and required or recommended account-security steps.

Cybersecurity lessons: Protect password hashes with modern schemes, limit access to account databases, monitor suspicious access, and disclose incidents promptly.

2. Equifax (2017)

Attack type: Web application exploitation / personal-data breach

What happened: Equifax announced a breach involving sensitive information of approximately 147 million people in the United States.

How the attack occurred: Attackers exploited a known Apache Struts vulnerability in a consumer dispute portal that had not been patched in time.

Impact: Exposed information included names, Social Security numbers, birth dates, addresses, and some driver's license and payment card details. The company faced settlements, response costs, and serious public concern.

Response and recovery: Equifax took the affected application offline, investigated, offered consumer support, and agreed to regulatory settlements.

Cybersecurity lessons: Maintain a complete asset inventory, patch critical vulnerabilities rapidly, validate fixes, and monitor data access.

3. Marriott International / Starwood (2018 disclosure)

Attack type: Long-running unauthorized access / guest database breach

What happened: Marriott announced unauthorized access to the Starwood guest reservation database; initial estimates were revised during investigation.

How the attack occurred: Attackers had gained access to the Starwood environment years before the 2018 discovery, illustrating risks in inherited systems following acquisitions.

Impact: Hundreds of millions of guest records were potentially affected; some records included passport information and reservation details. Exact categories differed among records.

Response and recovery: Marriott investigated, notified affected guests, set up support resources, and improved security oversight.

Cybersecurity lessons: Assess cybersecurity during mergers, monitor legacy environments, and minimize retention of sensitive guest data.

4. Sony Pictures Entertainment (2014)

Attack type: Destructive malware / data theft

What happened: An intrusion disrupted Sony Pictures systems and resulted in the publication of confidential company information.

How the attack occurred: Attackers penetrated corporate systems, stole internal information, and deployed destructive malware. Public US government attribution linked the attack to North Korea.

Impact: Employee data, emails, and unreleased material were exposed; workstations and business operations were disrupted.

Response and recovery: Sony rebuilt affected systems, worked with investigators, and strengthened security processes.

Cybersecurity lessons: Use offline backups, endpoint detection, least privilege, and separate critical systems to limit destructive attacks.

5. Colonial Pipeline (2021)

Attack type: Ransomware

What happened: Colonial Pipeline shut down its pipeline operations temporarily following a ransomware attack on its business IT network.

How the attack occurred: The DarkSide ransomware incident involved access through a compromised VPN account; reporting described an account without multifactor authentication.

Impact: The shutdown contributed to fuel supply disruptions and public concern across parts of the eastern United States. Colonial paid a ransom of about $4.4 million, part of which was later recovered by US authorities.

Response and recovery: The company isolated systems, suspended operations as a precaution, restored service, and coordinated with federal agencies.

Cybersecurity lessons: Require MFA for remote access, disable unused accounts, segment IT and operational technology, and practice incident-response shutdown decisions.

6. T-Mobile (2021)

Attack type: Unauthorized access / customer-data theft

What happened: T-Mobile disclosed a major breach involving information of current, former, and prospective customers.

How the attack occurred: An attacker accessed T-Mobile systems and extracted datasets. Public reporting and company statements described different affected groups and information categories.

Impact: Tens of millions of individuals were affected; exposed information in some datasets included names, dates of birth, Social Security numbers, and identification details.

Response and recovery: T-Mobile investigated, notified customers, offered identity-protection services, and announced additional security investments.

Cybersecurity lessons: Reduce stored sensitive data, apply strong access controls, monitor large data exports, and segment customer databases.

7. MGM Resorts International (2023)

Attack type: Social engineering / ransomware-related disruption

What happened: A cyber incident affected MGM hotel, casino, and digital services in September 2023.

How the attack occurred: Public reports associated the incident with social engineering and attackers connected to the Scattered Spider ecosystem; exact internal attack details are not all public.

Impact: MGM reported an estimated negative impact of roughly $100 million to adjusted property EBITDAR for the quarter, along with other expenses. Guest services and systems were disrupted.

Response and recovery: MGM shut down some systems, restored services over time, and engaged outside cybersecurity specialists.

Cybersecurity lessons: Harden help-desk identity verification, protect privileged account resets, and prepare manual operating procedures.

8. Change Healthcare (UnitedHealth Group) (2024)

Attack type: Ransomware / healthcare service interruption

What happened: A February 2024 attack disrupted Change Healthcare, a major US healthcare technology and claims-processing provider.

How the attack occurred: UnitedHealth executives stated attackers used compromised credentials to access a remote portal that lacked multifactor authentication. The ALPHV/BlackCat group was associated with the attack.

Impact: Claims processing and pharmacy-related workflows were disrupted nationwide. UnitedHealth later estimated that approximately 190 million individuals were affected by the data breach, subject to updates.

Response and recovery: The company disconnected affected services, rebuilt systems, provided financial assistance to providers, and undertook notification and recovery efforts.

Cybersecurity lessons: MFA must cover all remote entry points; highly connected service providers need continuity plans and carefully staged restoration.

9. Target (2013)

Attack type: Third-party access / point-of-sale malware

What happened: Target disclosed theft of payment card data during the 2013 holiday shopping season.

How the attack occurred: Attackers reportedly used credentials associated with an HVAC contractor to enter the environment and later installed malware on point-of-sale systems.

Impact: About 40 million payment card accounts were affected, while personal information related to up to 70 million individuals was also involved. These figures are different categories and should not simply be added together.

Response and recovery: Target investigated, replaced payment systems, notified customers, and expanded security controls.

Cybersecurity lessons: Restrict vendor privileges, separate payment networks, and act on security alerts quickly.

10. Microsoft corporate email intrusion (2024 disclosure)

Attack type: Password spraying / email espionage

What happened: Microsoft disclosed that a nation-state actor it calls Midnight Blizzard accessed a small percentage of corporate email accounts, including senior leaders and cybersecurity and legal staff.

How the attack occurred: Microsoft reported password-spray activity against a legacy non-production test tenant account, followed by use of permissions to access email.

Impact: Internal emails and attached documents were accessed. This was a corporate email compromise, not evidence that all Microsoft customer accounts were breached.

Response and recovery: Microsoft investigated, removed attacker access, notified affected parties, and described further security changes.

Cybersecurity lessons: Retire legacy identities, enforce MFA and conditional access, and minimize standing application permissions.

Common attack methods

Phishing and social engineering: Tricking employees or support desks into revealing credentials or changing account access.

Unpatched software: Exploiting a known vulnerability before an organization applies a fix.

Stolen credentials: Using valid passwords or tokens to bypass basic perimeter defenses.

Ransomware: Encrypting systems and/or stealing data to pressure an organization for payment.

Supply-chain and vendor access: Entering through a contractor, software provider, or acquired business system.

Data exfiltration: Copying sensitive records from databases, email, or file repositories.

Practical prevention checklist

• Use phishing-resistant MFA for administrators, remote access, and help-desk recovery.

• Keep an inventory of applications and patch critical vulnerabilities according to risk.

• Apply least privilege and review supplier and service-account permissions.

• Segment networks and monitor unusual data transfers and sign-ins.

• Encrypt sensitive information and retain only what is necessary.

• Maintain immutable/offline backups and regularly test restoration.

• Run incident-response drills and prepare customer and regulator notifications.

• Train employees on phishing, social engineering, and identity verification.

Conclusion

These incidents show that a cyberattack is not only a technical problem. It can interrupt essential services, expose personal information, create legal obligations, and damage a company's reputation. Prevention requires people, processes, and technology working together. A strong incident-response plan also helps limit harm when prevention fails.

References and further reading

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MGM Resorts, 2023 SEC filings: https://investors.mgmresorts.com/

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Microsoft Security Blog, Midnight Blizzard investigation (2024): https://www.microsoft.com/en-us/security/blog/

Note: Dates and counts reflect public disclosures and may change after later investigations. This report is for educational use, not a forensic finding.